

Bhanu Prakash Vangala

Doctoral Candidate, University of Missouri | Research funded by [NASA](#) | [DoD](#) | [NSF](#)

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Research Interests: LLM Systems and Serving | Agentic AI | Reproducibility and Provenance | Trustworthy AI | ML Evaluation

Education

University of Missouri, Columbia, MO

08/2023 – 05/2027 (expected)

Ph.D. in Computer Science, advised by Prof. Tanu Malik (Radiant Lab)

> Dissertation: *Executable Reliability: Unifying Trust, Reproducibility, and Efficiency in LLM-Based Systems*.

> [Google PhD Fellowship Nominee](#) (NLP track, 2025); EECS Graduate Travel Fellow (2026).

University of Missouri, Columbia, MO

08/2023 – 05/2025

M.S. in Computer Science

> Thesis: *Deploying LLMs as a Service in Kubernetes HPC Clusters*, advised by Prof. Grant J. Scott and Prof. Jianlin Cheng.

Vellore Institute of Technology, India

05/2019 – 04/2023

B.Tech. in Computer Science and Engineering (Data Analytics)

> Thesis: *Multilingual Sentiment Analysis on KOO User Posts*, with Prof. Ponnuram Kumaraguru and Dr. Soubhagya Barpanda.

Experience

Microsoft, Redmond, WA

05/2026 – 08/2026

Research Data Science Intern with Anqi Cheng and Arturo Herrera

> Built TempFE, a temporal feature framework for Windows retention telemetry; improved model PR-AUC by 30%.

> Designed a pre-registered six-arm benchmark in which the feature model outperformed TCN and TFT baselines.

University of Missouri, Columbia, MO

12/2023 – Present

Graduate Research Assistant with Tanu Malik (Radiant Lab) and Jianlin Cheng; funded by [NASA AIST](#), [NSF](#), and [DoD \(ERDC\)](#)

> Built cold-start-aware LLM routing (Thompson Sampling); cut GPU-hours 46% over 310K+ inference runs.

> Built a three-layer dependency audit (1,000 runs, 3 agents, 4 languages); agent agreement fell to 7%.

LearnLLM.dev, Remote

Present

Founder; built and maintain an LLM education platform with an active user base (Next.js, TypeScript, Supabase)

Precision and Automated Agriculture Lab (PAAL), Columbia, MO

08/2023 – 12/2023

Graduate Research Assistant, University of Missouri

> Built AI pipelines for UAV remote-sensing imagery; improved geospatial accuracy by 40% in UAV workflows.

Adobe Research, India

05/2022 – 01/2023

Volunteer Research Intern with Nandakishore Kambhatla; delivered quality-controlled scraping pipelines for research datasets

Publications

Published and Accepted

- [1] **One Threshold Does Not Fit All Languages: Language-Conditional Deferral for Reliable and Efficient Low-Resource Text Classification**
Bhanu Prakash Vangala, Navya Vangala
NeurIPS 2026 Workshop on Global South in AI (Accepted) (Oral Presentation) [GlobalSouthAI @ NeurIPS 2026]
- [2] **AI-Generated Code Is Not Reproducible (Yet): An Empirical Study of Environment Executability** 
Bhanu Prakash Vangala, A. Adibifar, Tanu Malik, Ashish Gehani
ACM Conference on Reproducibility and Replicability (Oral Presentation) [ACM REP 2026]
- [3] **Pick and Spin: Efficient Cold-Start-Aware Routing for Self-Hosted LLM Serving** 
Bhanu Prakash Vangala, Tanu Malik
IEEE International Conference on Cloud Computing (Oral Presentation) [IEEE CLOUD 2026]
- [4] **CAMP: Consumption-Aware Memory Prediction for Scientific Workflows** 
Bhanu Prakash Vangala, N. K., Tanu Malik
International Conference for High Performance Computing, Networking, Storage, and Analysis [SC 2026]
- [5] **Efficient Multi-Model Orchestration for Self-Hosted Large Language Models** 
Bhanu Prakash Vangala, Tanu Malik, Ashish Gehani
DAI Workshop at the AAAI Conference on Artificial Intelligence (Oral Presentation) [DAI @ AAAI 2026]

- [6] **HalluMat: A Benchmark Dataset and Framework for Hallucination Detection in LLMs for Materials Science** 
Bhanu Prakash Vangala, S. Mahmud, P. Neupane, J. Selvaraj, Jianlin Cheng
AAAI Conference on Artificial Intelligence (Oral Presentation) [AAAI 2025]
- [7] **HallFormer: A Transformer-Based Framework for Detecting Hallucination in Large Language Models** 
S. Mahmud, P. Neupane, J. Selvaraj, Bhanu Prakash Vangala, Jianlin Cheng
AAAI Conference on Artificial Intelligence (Oral Presentation) [AAAI 2025]

Under Review

- [8] **AdaptFlow: Efficient Adaptation of Accuracy-First AI Workflows under Data Drift**
Bhanu Prakash Vangala, Shankar Aditya, Todd Neif, Abhilash Jindal, Tanu Malik
Conference on Machine Learning and Systems (under review) [MLSys 2027]
- [9] **Code That Works, Environments That Don't: Measuring Environment Reproducibility in AI-Generated Software**
Bhanu Prakash Vangala, Tanu Malik
AI Magazine (AAAI, Wiley), Special Issue on AI and Reproducibility (under review) [AI Magazine 2026]
- [10] **How Many Labels Does a Language Need? Annotation Budgets and Cross-Lingual Pooling for African-Language Text Classification**
Bhanu Prakash Vangala, Sowmya Guda, Navya Vangala
NeurIPS 2026 Africa in AI Affinity Workshop (under review) [Africa in AI @ NeurIPS 2026]
- [11] **Evaluation Choices Shape Biomedical ML Claims: A Pediatric Pneumonia Benchmark Case Study**
Bhanu Prakash Vangala, Sowmya Guda, Latha Peddi, Navya Vangala
Workshop on Responsible Communication of ML Research in Biomedicine (under review) [RCMLR @ NeurIPS 2026]
- [12] **AI Sees, XAI Explains? Evaluating Explanation Reliability in Automated Seed Quality Inspection**
Bhanu Prakash Vangala, Navya Vangala
Computers and Electronics in Agriculture (under review) [Elsevier 2026]
- [13] **Visual Realism Does Not Predict Seed-Quality Utility in AI-Generated Synthetic Seeds**
Bhanu Prakash Vangala, Navya Vangala
Artificial Intelligence in Agriculture (under review) [Elsevier 2026]
- [14] **AI-Generated Seed Images Do Not Outperform Matched Reuse of Scarce Real Defect Images in Seed Quality Assessment**
Bhanu Prakash Vangala, Navya Vangala
Scientific Reports (under review) [Springer Nature 2026]
- [15] **Scores That Hold, Benchmarks That Leak: Measuring Dataset Contamination in Public Brain-Tumor MRI Classification**
Bhanu Prakash Vangala, Sowmya Guda, Latha Peddi, Navya Vangala
IEEE Journal of Biomedical and Health Informatics (under review) [IEEE JBHI 2026]

Posters and Presentations

- [16] **Adaptive Inference: Orchestrating Fine-Tuned LLMs with Serverless GPUs in HPC Environments** 
Bhanu Prakash Vangala, Tanu Malik
University of Missouri (Poster Presentation) [University of Missouri 2025]

Awards and Honors

Google PhD Fellowship Nominee	NLP track, University of Missouri	2025
Top 100, ACM Doctoral Summit	Association for Computing Machinery	2026
Chameleon Cloud Travel Award	Top 10 proposals, NSF Chameleon Cloud	2026
Outstanding Reviewer Award	NeurIPS	2025
EECS Graduate Travel Fellowship	University of Missouri	2026
Outstanding Master's Student Award	University of Missouri	2025
Runner-up, Generative AI for Social Good Hackathon	IBM, MUIDSI	2025
Dean's Research Excellence Award	Vellore Institute of Technology	2023

Teaching

Designing End-to-End ML Systems	Teaching Assistant; designed graded labs on Chameleon Cloud	Fall 2026
Web Development (MERN Stack)	Teaching Assistant; mentored 115+ students	Fall 2023 – Fall 2025

Service

Reviewer: NeurIPS 2025 ([Outstanding Reviewer](#)) and 2026, CIKM 2025, ACM CAIS 2026, IEEE, Elsevier journals.
Artifact Evaluation (AE): NeurIPS 2026, ACM CAIS 2026, ACM REP.